

COEP Technological University, Pune
Department of Computer Science and Engineering
MID SEMESTER EXAMINATION
Subject: Microprocessor Techniques

Duration: 90 minutes

Max. Marks: 30

1. A) Draw architecture of 8086, explain BIU and EU. State various addressing modes with one example each. No explanation required for addressing modes **(04 Marks)**

- B) Explain in comments field the execution of the program **(02 Marks)**

AX=89ABH, CS=6000H, DS=8000H, SP=FFFFH, SS:F000H, DI=5000H

CS: MOV [DI], AX

PUSH AX

POPF

- C) Explain in comments field execution details and number of bus cycles required

AX=089CH, CS=2000H, IP=0000H, BP=2000H, DS=1000, SS=3000 **(02 MARKS)**

CS:IP

OPCODE

MNEMONICS

2000:0000H

8B 46 FD

MOV [BP-03H], AX

2. A) Design a minimum mode 8086 CPU module with Clock and Power on Reset Circuit, separate Address and Data buses, separate necessary control signals for RAM and I/O. Make suitable assumptions wherever necessary. **(04 Marks)**

- B) Interface 32 Kbyte x 2 ROM and 32 Kbyte x 2 RAM to 8086. Starting address of ROM is F0000H and Starting address of RAM is 00000H **(04 Marks)**

- C) Write location and the instruction you would like to be executed by the processor on power up? Write a program to be executed as the first code upon reset, to initialize vector locations of NMI for following vector location address:

NMI: F000:0300H

(02 Marks)

- D) Write a main program to test NMI. Write interrupt service subroutine for NMI to increment the contents of memory location pointed by BX register. **(02 Marks)**

3. A) Interface 8255 to 8086 as an odd bank with starting address as F1H. Find the I/O port address map **(02 Marks)**

- B) Draw details of mode 1 input for 8255 Port A with $INTE_A = BSR$ of PC4. Conceptually correct diagram will be given full marks. Write program to demonstrate mode 1 input after initialization of 8255 in interrupt driven data transfer scheme. Explain the sequence of handshaking signals. **(03 Marks)**

4. Interface master 8259 to 8086 with base address of C0H, IR2 of the master is connected

with slave 8259 with base address of 81H. Base type number for Master is 80H and for slave F0H. Design for non-buffered cascaded mode. IRO of master has vector location address F000:2000H. IR4 of slave has vector location address F000:5000H. Write Interrupt Service Subroutine for Master 8259's IRO level, to decrement contents of memory pointed by BX. Write Interrupt Service Subroutine for IR4 of slave to decrement contents of memory location pointed by BX by two. Ensure normal fully nested mode. **(05 Marks)**

ADDITIONAL DATA FOR QUESTION 4:

ICW1: A0=0

X	X	X	1	LTIM	ADI	SNGL	IC4
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ICW2: A0=1

T7	T6	T5	T4	T3	X	X	X
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ICW3 (MASTER): A0=1

S7	S6	S5	S4	S3	S2	S1	S0
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ICW3 (SLAVE): A0=1

0	0	0	0	0	ID2	ID1	ID0
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ICW4: (A0=1)

0	0	0	SFNM	BUF	M/S	AEOI	μPM
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OCW1: (A0=1)

M7	M6	M5	M4	M3	M2	M1	M0
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OCW2: (A0=0)

R	SL	EOI	0	0	L2	L1	L0
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R	SL	EOI	
0	0	1	NON-SPECIFIC EOI
0	1	1	SPECIFIC EOI
1	0	1	ROTATE ON NON-SPECIFIC EOI
1	0	0	ROTATE IN AUTOMATIC EOI (SET)
0	0	0	ROTATE IN AUTOMATIC EOI (CLEAR)
1	1	1	ROTATE ON SPECIFIC EOI
1	1	0	SET PRIORITY

L2	L1	L0	IR LEVEL TO BE ACTED UPON
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